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GENETICALLY MODIFIED YEAST AND FERMENTATION PROCESSES FOR THE PRODUCTION OF ARABITOL

Abstract

Disclosed herein are genetically engineered yeast cells capable of producing arabitol. The engineered yeast cell may comprise an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs: 1, 2, 3, 9, or 11.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/364,359, filed May 9, 2022, which is incorporated by reference herein in its entirety.

REFERENCE TO A SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

[0002] The content of the Sequence Listing XML file of the sequence listing named “PT-1351-WO-PCT.xml” which is 150,554 bytes in size created on May 4, 2023 and electronically submitted via Patent Center herewith the application is incorporated by reference in its entirety.

BACKGROUND

[0003] Xylitol is a low-calorie sweetener used as a food additive and sugar substitute. Commonly used in drug, dietary supplement, confectionary, and toothpaste compositions, xylitol has also been associated with anticariogenic properties when used in chewing gums. Traditional methods of xylitol production, including chemically catalyzed hydrogenation of xylose hydrolyzed from biomass extracted xylan, are both monetarily and environmentally costly. These methods require high temperatures and pressures, large amounts of water, and metal catalysts that must be mined. In contrast, fermentation processes have been used commercially at large scale to produce other organic molecules, such as ethanol, citric acid, lactic acid, and the like, and may offer a cost effective and sustainable alternative to traditional xylitol processing methods.

[0004] In the development of microorganism-based fermentation strategies for the production of xylitol, production of metabolic pathway intermediates and alternative fermentation products are important considerations. For example, metabolic pathways active in the production of xylitol may have overlap with the metabolic pathways for the production of arabitol, erythritol, ribitol, and the like. The intermediates and products have their own uses and markets that make their fermentation commercially relevant. Accordingly, provided herein are genetically modified yeast and fermentation methods for the production of arabitol.

SUMMARY

[0005] The present disclosure provides a genetically engineered yeast cell capable of producing arabitol, the engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11. The yeast cell may be an osmotolerant yeast cell. The yeast cell may be a cell of the subphylum Ustilaginomycotina. The yeast cell may be selected from the group consisting of *Trichosporonoides megachiliensis*, *Trychosporonoides oedocephalis*, *Trychosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*. The yeast cell may be a yeast cell of the genus.

[0006] The disclosure also provides a genetically engineered *Moniliella* cell capable of producing arabitol, the engineered *Moniliella* cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.

[0007] The ARD2DH enzyme may have a sequence at least 85% identical to SEQ ID NO:2, 3, 9,

and/or 11. The ARD2DH enzyme may have a sequence at least 90% identical to SEQ ID NO: 2, 3, 9, and/or 11.

[0008] The engineered cell described herein may be a *Moniliella pollinis* cell. The yeast cell described herein may be capable of producing arabitol at a titer of at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L when used in a fermentation process in the presence of dextrose at 35° C. for 96 hours. Erythritol production by the engineered cell described herein may be reduced relative to erythritol production in an equivalent yeast cell lacking the exogenous polynucleotide sequence.

[0009] The exogenous polynucleotide sequence may be operably linked to a heterologous or artificial promoter. The promoter may be a constitutive promoter. The promoter may be selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

[0010] The disclosure also provides a method for producing arabitol using the engineered cells described herein, the method comprising contacting a substrate comprising dextrose with an engineered cell described herein, wherein fermentation of the substrate by the engineered yeast produces arabitol. The disclosure also provides a method for producing arabitol, the method comprising contacting a substrate comprising dextrose with an engineered yeast cell comprising an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogenase (ARD2DH) enzyme comprising a sequence at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98%, at least 99%, or 100% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11, wherein fermentation of the substrate by the engineered yeast produces arabitol. The fermentation temperature may be at or between 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C. The volumetric oxygen uptake rate (OUR) may be between 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L.Math.h). Erythritol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Erythritol production may be less than 60, 50, 40, or less than 30 g/L when the fermentation is run at 35° C. for 96 hours. Arabitol production may be at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L when the fermentation is run at 35° C. for 96 hours. Glycerol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence. Ethanol production may be reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] This patent or application contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawings will be provided by the Office upon request and the payment of the necessary fee.

[0012] The drawings illustrate generally, by way of example, but not by way of limitation, various aspects discussed herein.

[0013] FIG. 1 shows the predicted native pentose phosphate pathway (dotted lines and arrows) and the native glycolysis pathways (solid lines and arrows) in *Moniliella pollinis*.

[0014] FIG. 2 shows diversity in the arabitol 2-dehydrogenase (ARD2DH) sequence space.

[0015] FIG. 3 shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask

fermentation as outlined in Example 3. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0016] FIG. 4 shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 3. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0017] FIG. 5 shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 4. The dotted line shows the level of arabitol production in the parent strain 1-1.

[0018] FIG. 6 shows arabitol concentrations (g/L) at 72 hours and 96 hours of shake flask fermentation as outlined in Example 4. The dotted line shows the level of arabitol production in the parent strain 1-1.

DETAILED DESCRIPTION

[0019] Reference will now be made in detail to certain aspects of the disclosed subject matter, examples of which are illustrated in part in the accompanying drawings. While the disclosed subject matter will be described in conjunction with the enumerated claims, it will be understood that the exemplified subject matter is not intended to limit the claims to the disclosed subject matter.

[0020] In this document, the terms “a,” “an,” or “the” are used to include one or more than one unless the context clearly dictates otherwise. The term “or” is used to refer to a nonexclusive “or” unless otherwise indicated. All publications, patents, and patent documents referred to in this document are incorporated by reference herein in their entirety, as though individually incorporated by reference. In the event of inconsistent usages between this document and those documents so incorporated by reference, the usage in the incorporated reference should be considered supplementary to that of this document; for irreconcilable inconsistencies, the usage in this document controls.

[0021] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include all the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range were explicitly recited. For example, a range of “about 0.1% to about 5%” or “about 0.1% to 5%” should be interpreted to include not just about 0.1% to about 5%, but also the individual values (e.g., 1%, 2%, 3%, and 4%) and the sub-ranges (e.g., 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement “about X to Y” has the same meaning as “about X to about Y,” unless indicated otherwise. Likewise, the statement “about X, Y, or about Z” has the same meaning as “about X, about Y, or about Z,” unless indicated otherwise.

[0022] Unless expressly stated, ppm (parts per million), percentage, and ratios are on a by weight basis. Percentage on a by weight basis is also referred to as wt % or % (wt) below.

[0023] This disclosure relates to various recombinant cells engineered to produce arabitol. In general, the recombinant cells described herein have an active pentose phosphate pathway and are characterized by expression of an exogenous arabitol 2-dehydrogenase (ARD2DH) enzyme. The disclosure further provides fermentation methods for the production of arabitol from dextrose using the genetically engineered cells described herein.

[0024] In general, recombinant cells described herein are yeast cells. As used herein, “yeast” refers to eukaryotic single celled microorganisms classified as members of the fungus kingdom. Yeast are unicellular organisms which evolved from multicellular ancestors with some species retaining multicellular characteristics such as forming strings of connected budding cells known as pseudo hyphae or false hyphae. Yeast cells may also be referred to in the art as yeast-like cells, and as used herein “yeast cell” encompasses both yeast and yeast-like cells. Suitable yeast and yeast-like host cells for modification may include, but are not limited to, *Saccharomyces cerevisiae*, *Komagataella* sp., *Kluyveromyces* (e.g., *Kluyveromyces lactis*, *Kluyveromyces marxianus*), *Yarrowia lipolytica*, *Issatchenkia orientalis*, *Pichia galeiformis*, *Pichia* sp. YB-4149 (NRRL designation), *Pichia*

pastoris, *Candida* (e.g., *Candida magnoliiae*, *Candida deserticola*, *Pichia membranifadens*, *Pichia fermentans*, *Aspergillus*, *Trichoderma*, *Myceliphthora thermophila*, *Moniliella* (e.g., *Moniliella pollinis*), *Pfaffia*, *Yamadazyma*, *Hansenula*, *Pichia kudriavzevii*, *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trychosporonoides oedocephalis*, *Trychosporonoides nigrescens*), *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Penicillium*, and *Torula*. An ordinarily skilled artisan would understand the requirements for selection of a suitable yeast cell, and recombinant yeast cells of the present disclosure are not limited to those expressly recited herein. Methods for genetic engineering of yeast cells are known and described in the art and a skilled artisan would understand the methods necessary to transform and engineer a suitable yeast cell.

[0025] A suitable yeast cell may be a cell of the phylum Basidiomycota and the subphylum Ustilaginomycotina. Suitable yeast of the subphylum Ustilaginomycotina include, but are not limited to, *Ustilago* (e.g., *U. cynodontis*, *U. maydis*, *U. sphaerogena*, *U. cordal*, *U. scitaminea*, *U. coicis*, *U. syntherismae*, *U. esculenta*, *U. neglecta*, *U. crus-galli*, *Ustilago avenae*), *Sporisorium* (e.g., *Sporisorium exsertum*), *Moniliella* (e.g., *M. pollinis*, *M. tomentosa*, *M. acetoabutans*, *M. fonsecae*, *M. madida*, *M. megachiliensis*, *M. oedocephalis*, *M. nigrescens*), and *Pseudozyma* (e.g., *Pseudozyma tsukubaensis*), and *Trichosporonoides* (e.g., *Trichosporonoides megachiliensis*, *Trychosporonoides oedocephalis*, *Trychosporonoides nigrescens*). Yeast of the subphylum Ustilaginomycotina have been known and described in the art as potential production organisms for valuable chemicals such as itaconate, malate, succinate, mannitol, and erythritol and other valuable biotechnological applications. See, for example, Geiser et al. (Prospecting the biodiversity of the fungal family Ustilaginaceae for the production of value-added chemicals,” *Fungal Biol Biotechnol*, 2014, 1:2), Feldbrugge et al., (“The biotechnological use and potential of plant pathogenic smut fungi,” *Appl Microbiol Biotechnol*, 2013, 97(8):3253-65), Guevarra et al., (“Accumulation of itaconic, 2-hydroxyparaconic, itatartaric, and malic acids by strains of the genus *Ustilago*, *Agric. Biol. Chem.*, 1990, 54(9), 2353-2358), and Moon et al., (“Biotechnological production of erythritol and its applications,” *Appl Microbiol Biotechnol*, 2010, 86:1017-1025).

[0026] A suitable yeast cell will have an active pentose phosphate pathway that produces ribulose-5-phosphate. As used herein “active pentose phosphate pathway” refers to expression of one or more functional enzymes which, together, convert glucose-6-phosphate, NADP^{sup.+} or NAD⁺, and water to NADPH or NADH, CO₂, and ribulose-5-phosphate. Continuing in a non-oxidative phase, the pathway may also produce other pentose (i.e., 5-carbon) sugars. For example, the pentose phosphate pathway may produce ribulose-5-phosphate, ribose-5-phosphate, xylulose-5-phosphate, fructose 6-phosphate, combinations thereof, and the like, depending on the enzymatic activities present. The active pentose phosphate pathway may be native to the yeast cell or it may be introduced into the yeast cell by genetic engineering.

[0027] The yeast cell may be an osmotolerant yeast cell. As used herein, “osmotolerant” refers to a yeast capable of growth and reproduction under conditions of high osmolarity, such as at least 10% (w/v), at least 20% (w/v), at least 30% (w/v), at least 40% (w/v), at least 50% (w/v), or at least 60% (w/v) glucose and/or at least 6% (w/v), at least 10% (w/v), at least 12% (w/v), at least 13% (w/v), at least 15% (w/v) sodium chloride. Species and strains of osmotolerant yeast are known and described in the art, including many species of yeast used in industrial fermentation processes. Likewise, methods for assaying yeast osmotolerance are known and described in the art. See, for example, Tiwari, S., et al., (“Nectar yeast community of tropical flowering plants and assessment of their osmotolerance and xylitol-producing potential,” *Current Microbiology*, 2022, 79:28).

[0028] The recombinant yeast cell may be a recombinant *Moniliella* cell, for example, a *Moniliella pollinis* cell. FIG. 1 shows the predicted native pentose phosphate and glycolysis pathways in *Moniliella pollinis*. *Moniliella* has previously been used in the fermentation production of erythritol and methods for genetically modifying and fermenting *Moniliella* are known and described in the art. See, for example, Li et al. (“Methods for genetic transformation of filamentous fungi,” 2017,

Microb Cell Fact, 16:168).

[0029] Various plasmids and methods for transformation of *Moniliella* are also described in the Examples below. For example, *Moniliella* may be transformed using a bipartite polynucleotide sequence(s) in which, following recombination, the exogenous polynucleotide of interest is integrated at the specified locus and the selection marker is expressible within the cell. Suitable selection markers are known and used in the art. The selectable marker may include, but is not limited to, amdS (for example broken into a 3' portion, SEQ ID NO:63, and a 5' portion, SEQ ID NO:64), G418 resistance gene (for example broken into a 3' portion, SEQ ID NO:69, and a 5' portion, SEQ ID NO:70), zeocin resistance gene (for example broken into a 3' portion, SEQ ID NO:65, and a 5' portion, SEQ ID NO:66), nourseothricin N-acetyl transferase (NAT) (for example broken into a 3' portion, SEQ ID NO:67, and a 5' portion, SEQ ID NO:68), and invertase gene (SUC2) (for example a 3' portion of SEQ ID NO:71 and a 5' portion of SEQ ID NO:72).

[0030] The recombinant cells described herein include one or more exogenous polynucleotide sequences encoding one or more exogenous polypeptides that, when expressed improve the fermentation of glucose to ribitol by the recombinant cells.

[0031] The terms “glucose” and “dextrose” are used interchangeably herein and refer to D-glucose except where expressly indicated otherwise.

[0032] As used herein, “exogenous” refers to genetic material or an expression product thereof that originates from outside of the host organism. For example, the exogenous genetic material or expression product thereof can be a modified form of genetic material native to the host organism, it can be derived from another organism, it can be a modified form of a component derived from another organism, or it can be a synthetically derived component. For example, a *K. lactis* invertase gene is exogenous when introduced into *S. cerevisiae*.

[0033] As used herein, “native” refers to genetic material or an expression product thereof that is found, apart from individual-to-individual mutations which do not affect function or expression, within the genome of wild-type cells of the host cell. For the purposes of this application, the *Moniliella pollinis* cell “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUCL40385, is considered the wild-type *Moniliella pollinis* cell.

[0034] As used herein, the terms “polypeptide” and “peptide” are used interchangeably and refer to the collective primary, secondary, tertiary, and quaternary amino acid sequences and structure necessary to give the recited macromolecule its function and properties. As used herein, “enzyme” or “biosynthetic pathway enzyme” refer to a protein that catalyzes a chemical reaction. The recitation of any particular enzyme, either independently or as part of a biosynthetic pathway is understood to include the co-factors, co-enzymes, and metals necessary for the enzyme to properly function. A summary of the amino acids and their three and one letter symbols as understood in the art is presented in Table 1. The amino acid name, three letter symbol, and one letter symbol are used interchangeably herein.

TABLE-US-00001 TABLE 1 Amino Acid three and one letter symbols

Amino Acid	Three letter symbol	One letter symbol
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic acid	Asp	D
Cysteine	Cys	C
Glutamic acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

[0035] Variants or sequences having substantial identity or homology with the polypeptides described herein can be utilized in the practice of the disclosed recombinant cells, compositions, and methods. Such sequences can be referred to as variants or modified sequences. That is, a polypeptide sequence can be modified yet still retain the ability to exhibit the desired activity.

Generally, the variant or modified sequence may include greater than about 45%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95% sequence identity with the wild-type, naturally occurring polypeptide sequence, or with a variant polypeptide as described herein.

[0036] As used herein, the phrases “% sequence identity,” “% identity,” and “percent identity,” are used interchangeably and refer to the percentage of residue matches between at least two amino acid sequences or at least two nucleic acid sequences aligned using a standardized algorithm. Methods of amino acid and nucleic acid sequence alignment are well-known. Sequence alignment and generation of sequence identity include global alignments and local alignments which are carried out using computational approaches. An alignment can be performed using BLAST (National Center for Biological Information (NCBI) Basic Local Alignment Search Tool) version 2.2.31 software with default parameters. Amino acid % sequence identity between amino acid sequences can be determined using standard protein BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 6; Max matches in a query range: 0; Matrix: BLOSUM62; Gap Costs: (Existence: 11, Extension: 1); Compositional adjustments: Conditional compositional score matrix adjustment; Filter: none selected; Mask: none selected. Nucleic acid % sequence identity between nucleic acid sequences can be determined using standard nucleotide BLAST with the following default parameters: Max target sequences: 100; Short queries: Automatically adjust parameters for short input sequences; Expect threshold: 10; Word size: 28; Max matches in a query range: 0; Match/Mismatch Scores: 1, -2; Gap costs: Linear; Filter: Low complexity regions; Mask: Mask for lookup table only. A sequence having an identity score of XX % (for example, 80%) with regard to a reference sequence using the NCBI BLAST version 2.2.31 algorithm with default parameters is considered to be at least XX % identical or, equivalently, have XX % sequence identity to the reference sequence.

[0037] Polypeptide or polynucleotide sequence identity may be measured over the length of an entire defined polypeptide sequence, for example, as defined by a particular SEQ ID number, or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length supported by the sequences shown herein, in the tables, figures or Sequence Listing, may be used to describe a length over which percentage identity may be measured.

[0038] The polypeptides disclosed herein may include “variant” polypeptides, “mutants,” and “derivatives thereof.” As used herein the term “wild-type” is a term of the art understood by skilled persons and means the typical form of a polypeptide as it occurs in nature as distinguished from variant or mutant forms. As used herein, a “variant,” “mutant,” or “derivative” refers to a polypeptide molecule having an amino acid sequence that differs from a reference protein or polypeptide molecule. A variant or mutant may have one or more insertions, deletions, or substitutions of an amino acid residue relative to a reference molecule.

[0039] The amino acid sequences of the polypeptide variants, mutants, derivatives, or fragments as contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant, mutant, derivative, or fragment polypeptide may include conservative amino acid substitutions relative to a reference molecule. “Conservative amino acid substitutions” are those substitutions that are a substitution of an amino acid for a different amino acid where the substitution is predicted to interfere least with the properties of the reference polypeptide. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference polypeptide. Conservative amino acid substitutions generally maintain (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a beta sheet or alpha helical conformation, (b) the charge and/or hydrophobicity of the molecule at the site of the substitution, and/or (c) the bulk of the side chain.

[0040] As used herein, terms “polynucleotide,” “polynucleotide sequence,” and “nucleic acid sequence,” and “nucleic acid,” are used interchangeably and refer to a sequence of nucleotides or any fragment thereof. These phrases also refer to DNA or RNA of natural or synthetic origin, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. The DNA polynucleotides may be a cDNA (e.g., coding DNA) or a genomic DNA sequence (e.g., including both introns and exons).

[0041] A polynucleotide is said to encode a polypeptide if, in its native state or when manipulated by methods known to those skilled in the art, it can be transcribed and/or translated to produce the polypeptide or a fragment thereof. The anti-sense strand of such a polynucleotide is also said to encode the sequence.

[0042] Those of skill in the art understand the degeneracy of the genetic code and that a variety of polynucleotides can encode the same polypeptide. In some aspects, the polynucleotides (i.e., polynucleotides encoding an ARD2DH polypeptide) may be codon-optimized for expression in a particular cell including, without limitation, a plant cell, bacterial cell, fungal cell, or animal cell. While polypeptides encoded by polynucleotide sequences found in various species are disclosed herein any polynucleotide sequences may be used which encodes a desired form of the polypeptides described herein. Thus, non-naturally occurring sequences may be used. These may be desirable, for example, to enhance expression in heterologous expression systems of polypeptides or proteins. Computer programs for generating degenerate coding sequences are available and can be used for this purpose. Pencil, paper, the genetic code, and a human hand can also be used to generate degenerate coding sequences.

[0043] The recombinant cells described herein may include deletions or disruptions in one or more native genes. The phrase “deletion or disruption” refers to the status of a native gene in the recombinant cell that has either a completely eliminated coding region (deletion) or a modification of the gene, its promoter, or its terminator (such as by a deletion, insertion, or mutation) so that the gene no longer produces an active expression product, produces severely reduced quantities of the expression product (e.g., at least a 75% reduction or at least a 90% reduction) or produces an expression product with severely reduced activity (e.g., at least 75% reduced or at least 90% reduced). The deletion or disruption can be achieved by genetic engineering methods, forced evolution, mutagenesis, RNA interference (RNAi), and/or selection and screening. The native gene to be deleted or disrupted may be replaced with an exogenous nucleic acid of interest for the expression of an exogenous gene product (e.g., polypeptide, enzyme, and the like).

[0044] The recombinant cells described herein may include one or more genetic modifications in which an exogenous nucleic acid is integrated into the genome of the host cell. One of skill in the art know how to select suitable loci in a yeast genome for integration of the exogenous nucleic acid. Suitable integration loci may include, but are not limited to, the PDC1, GPD1, CYB2A, CYB2B, g4240, YMR226, MDHB, ATO2, Adh9091, Adh1202, ADE2, ADH2556, GAL6, MDH1, SCW11, ER1, ER3, pyrF, TRP3, gpdIIA, and gpdIIB loci. For example, in a *M. pollinis* host cells, suitable interaction loci may include, but are not limited to, the ER1 locus (defined as the locus flanked by SEQ ID NO:36 and SEQ ID NO:37), the ER3 locus (defined as the locus flanked by SEQ ID NO:24 and SEQ ID NO:25), the PDC1 locus (defined as the locus flanked by SEQ ID NO:26 and SEQ ID NO:27), the pyrF locus (defined as the locus flanked by SEQ ID NO:28 and SEQ ID NO:29), the TRP3 locus (defined as the locus flanked by SEQ ID NO:32 and SEQ ID NO:33), the gpdIIA locus (defined as the locus flanked by SEQ ID NO:34 and SEQ ID NO:35); and the gpdIIB locus (defined as the locus flanked by SEQ ID NO:38 and SEQ ID NO:39). Other suitable integration loci may be determined one of skill in the art. Furthermore, one of skill in the art would recognize how to use sequences to design primers to verify correct gene integration at the chosen locus.

[0045] The recombinant cell may have one or more copies of a given exogenous nucleic acid sequence integrated in a host chromosome(s) and replicated together with the chromosome(s) into

which it has been integrated. For example, the yeast cell may be transformed with nucleic acid construct including a polynucleotide sequence encoding for a polypeptide described herein and the polynucleotide sequence encoding for the polypeptide may be integrated in one or more copies in a host chromosome(s). The recombinant cell may include multiple copies (two or more) of a given polynucleotide sequence encoding a polypeptide described herein. The recombinant cell may have one, two, three, four, five, six, seven, eight, nine, ten, or more copies of a polynucleotide sequence encoding a polypeptide described herein integrated into the genome. The multiple copies of said polynucleotide sequence may all be incorporated at a single locus or may be incorporated at multiple loci.

[0046] The recombinant cells described herein are capable of producing arabinol and include an exogenous polynucleotide sequence encoding an arabinol 2-dehydrogenase (ARD2DH) enzyme. The exogenous polynucleotide sequence may be an exogenous ARD2DH gene.

[0047] An “arabinol 2-dehydrogenase gene” and an “ARD2DH gene” are used interchangeably herein and refer to any gene or polynucleotide that encodes a polypeptide with arabinol 2-dehydrogenase activity. As used herein “arabinol 2-dehydrogenase activity” refers to the ability to catalyze the conversion of D-ribulose and NADH or NADPH to D-arabinol and NAD^{sup.+} or NADP^{sup.+}. Enzymes with arabinol 2-dehydrogenase may be characterized under Enzyme Classification 1.1.1.250. The ARD2DH gene may be derived from any suitable source. For example, the ARD2DH gene may be derived from *Beauveria bassiana*, *Pichia stipitis*, *Candida albicans*, *Kwoniella heveanensis*, *Candida maltosa*. The ARD2DH gene may encode a polypeptide with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:1, 2, 3, 9, or 11. The ARD2DH gene may encode a polypeptide with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of at least one of SEQ ID NOs:2, 3, 9, or 11.

[0048] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Beauveria bassiana* ARD2DH gene encoding the amino acid of SEQ ID NO:1. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:1.

[0049] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Pichia stipitis* ARD2DH gene encoding the amino acid of SEQ ID NO:2. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:2.

[0050] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Candida albicans* ARD2DH gene encoding the amino acid of SEQ ID NO:3. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:3.

[0051] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Kwoniella heveanensis* ARD2DH gene encoding the amino acid of SEQ ID NO:9. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:9.

[0052] The recombinant cell may comprise an exogenous polynucleotide that is or may be derived from a *Candida maltosa* ARD2DH gene encoding the amino acid of SEQ ID NO:11. The exogenous polynucleotide may encode an amino acid sequence with at least 50%, at least 60%, at least 70%, at least 80%, at least 85%, at least 90%, at least 95%, at least 97%, or at least 99% sequence identity to the amino acid sequence of SEQ ID NO:11.

[0053] The exogenous polynucleotides in the recombinant cells described herein may be under the control of a promoter. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial promoter. Suitable promoters are known and described in the art. Promoters may include, but are not limited to, pyruvate decarboxylase promoter (PDC), translation elongation factor 2 promoter (TEF2), SED1, alcohol dehydrogenase 1A promoter (ADH1), hexokinase 2 promoter (HXK2), FLO5 promoter, pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

[0054] The exogenous nucleic acids in the recombinant cells described herein may be under the control of a terminator. For example, the exogenous nucleic acid may be operably linked to a heterologous or artificial terminator. Suitable terminators are known and described in the art. Terminators may include, but are not limited to, GAL10 terminator, PDC terminator, transaldolase terminator (TAL) 6PGD terminator (6PGDt; SEQ ID NO:51); ASNS terminator (ASNSt; SEQ ID NO:52); ENO1 terminator (ENO1t; SEQ ID NO:53); hexokinase 1 terminator (HXK1t; SEQ ID NO:54); PGK1 terminator (PGK1t; SEQ ID NO:55); PGM1 terminator (PGM1t; SEQ ID NO:56); PYK1 terminator (PYK1t; SEQ ID NO:57); RPLA terminator (RPLAt; SEQ ID NO:58); transaldolase 1 terminator (TAL1t; SEQ ID NO:59); TDH3 terminator (TDH3t; SEQ ID NO:60); translation elongation factor 2 terminator (TEF2t; SEQ ID NO:61); and triosephosphate isomerase 1 terminator (TPI1t; SEQ ID NO:62).

[0055] A promoter or terminator is “operably linked” to a given polynucleotide (e.g., a gene) if its position in the genome or expression cassette relative to said polynucleotide is such that the promoter or terminator, as the case may be, performs its transcriptional control function.

[0056] The polypeptides described herein may be provided as part of a construct. As used herein, the term “construct” refers to recombinant polynucleotides including, without limitation, DNA and RNA, which may be single-stranded or double-stranded and may represent the sense or the antisense strand. Recombinant polynucleotides are polynucleotides formed by laboratory methods that include polynucleotide sequences derived from at least two different natural sources or they may be synthetic. Constructs thus may include new modifications to endogenous genes introduced by, for example, genome editing technologies. Constructs may also include recombinant polynucleotides created using, for example, recombinant DNA methodologies. The construct may be a vector including a promoter operably linked to the polynucleotide encoding a polypeptide as described herein. As used herein, the term “vector” refers to a polynucleotide capable of transporting another polynucleotide to which it has been linked. The vector may be a plasmid, which refers to a circular double-stranded DNA loop into which additional DNA segments may be integrated.

[0057] The disclosure also provides fermentation methods for the production of arabitol using the recombinant cells described herein. The fermentation methods include the step of fermenting a substrate using the genetically engineered yeasts described herein to product arabitol. The fermentation method can include additional steps, as would be understood by a person skilled in the art. Non-limiting examples of additional process steps include maintaining the temperature of the fermentation broth within a predetermined range, adjusting the pH during fermentation, and isolating the arabitol from the fermentation broth. The fermentation process may be a fully aerobic process.

[0058] The fermentation method can be run using a suitable fermentation substrate. The substrate of the fermentation method can include glucose, sucrose, galactose, mannose, molasses, xylose,

fructose, hydrolysates of starch, lignocellulosic hydrolysates, or a combination thereof. One skilled in the art will recognize what fermentation substrate is suitable for a given fermentation organism and system.

[0059] The fermentation process can be run under various conditions. The fermentation temperature, i.e., the temperature of the fermentation broth during processing, may be ambient temperature. Alternatively, or additionally, the fermentation temperature may be maintained within a predetermined range. For example, the fermentation temperature can be maintained in the range of 25° C. to 45° C., 30° C. to 40° C., or 32° C. to 37° C., preferably about 35° C. However, a skilled artisan will recognize that the fermentation temperature is not limited to any specific range or temperature recited herein and may be modified as appropriate.

[0060] The fermentation process can be run within certain oxygen uptake rate (OUR) ranges. The volumetric OUR of the fermentation process can be in the range of 0.5 to 40, 1 to 35, 2 to 30, 3 to 25, 4 to 20, or 5 to 15 mmol O₂/(L.Math.h). In some embodiments, the specific OUR can be in the range of 0.05 to 10, 0.1 to 8, 0.15 to 5, 0.2 to 1, or 0.3 to 0.75 mmol O₂/(g cell dry weight.Math.h). However, the volumetric or specific OURs of the fermentation process are not limited to any specific rates or ranges recited herein.

[0061] The fermentation process can be run at various cell concentrations. In some embodiments, the cell dry weight at the end of fermentation can be 5 to 40, 8 to 30, or 10 to 20 g cell dry weight/L. Further, the pitch density or pitching rate of the fermentation process can vary. In some embodiments, the pitch density can be 0.05 to 11, 0.1 to 10, or 0.25 to 8 g cell dry weight/L.

[0062] The initial dextrose concentration of the fermentation may be at least 100, 200, 250, 300, 350, or at least 400 g/L dextrose. The initial dextrose concentration may be between 100 to 400, 150 to 350, or 250 to 325 g/L.

[0063] The fermentation process can be associated with various characteristics, such as, but not limited to, fermentation production rate, pathway fermentation yield, final titer, and peak fermentation rate. These characteristics can be affected by the selection of the yeast and/or genetic modification of the yeast used in the fermentation process. These characteristics can be affected by adjusting the fermentation process conditions. These characteristics can be adjusted via a combination of yeast selection or modification and the selection of fermentation process conditions.

[0064] The final arabitol titer of the process may be at least 0.2, 0.5, 0.75, 1.0, 1.5, or 2.0 g/L.

[0065] The fermentation process can be run as a dextrose-fed batch. Further, the fermentation process can be a batch process, continuous process, or semi-continuous process, as would be understood by a person skilled in the art.

EXAMPLES

[0066] The invention is further described in detail by reference to the following experimental examples. These examples are provided for purposes of illustration only and are not intended to be limiting unless otherwise specified. Thus, the invention should in no way be construed as being limited to the following examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

Example 1—D-Arabitol 2-Dehydrogenase Diversity

[0067] 509 candidate D-arabitol 2-dehydrogenase (ARD2DH) enzyme sequences were selected from the Uniprot database and analyzed. FIG. 2 demonstrates the sequences diversity for this set of sequences. This set is diverse, with 35% of the sequences having no homolog with more than 75% sequence identity. Eleven of these sequences were chosen for further characterization in vivo.

Example 2—Genetically Modified *Moniliella pollinis* Strains

[0068] Strain 1-1 is the *Moniliella pollinis* host strain “*Moniliella tomentosa* var *pollinis* TCV364” described in U.S. Pat. No. 6,440,712, which is incorporated herein by reference in its entirety, and deposited under the Budapest Treaty at BCCM/MUCL (Belgian Coordinated Collections of Micro-organisms/Mycothèque de l'Université Catholique de Louvain by *Eridania* Béghin Say, Vilvoorde

R&D Centre, Havenstraat 84, B-1800 Vilvoorde) on Mar. 28, 1997 under number MUC140385. Table 2 below lists various *Moniliella pollinis* strains, including information on the parent strain, the sequence with which the parent strain was transformed, and characterizations of the expression cassette(s) contained on the transformed sequence. Each “ARD2DH Homolog Expression Cassette” contained, in order, a 3' Zeocin resistance gene bipartite fragment (SEQ ID NO:65), the TEF2 terminator (SEQ ID NO:61), an Mp6PGD promoter (SEQ ID NO:40), a gene encoding the indicated ARD2DH homolog (one of SEQ ID NOs:1-11), an Mp6PGD terminator (SEQ ID NO:51), and a 3' ER3 flanking sequence (SEQ ID NO:25). Each “Selectable Marker Cassette” contained, in order, a 5' ER3 flanking sequence (SEQ ID NO:24), and a 5' Zeocin resistance gene bipartite fragment (SEQ ID NO:66).

[0069] Upon bipartite transformation with both the ARD2DH Homolog Expression Cassette and the Selectable Marker Cassette, the two cassettes recombine for integration of both the nucleotide sequence encoding the ARD2DH homolog and the Zeocin selectable marker at the ER3 locus.

[0070] The indicated *Moniliella pollinis* parent strain was transformed with the indicated sequence(s) by first protoplasting the parent strain by adding an enzyme mixture containing 0.6M MgSO₄, 7.5 g/L driselase, and 12.5 g/L *Trichoderma harzianum* lysing enzyme to a mycelial pellet of the parent strain. Protoplasts were then pelleted, washed with 0.6M MgSO₄, and resuspended in STC medium (0.6M sucrose, 50 mM CaCl₂, 10 mM Tris-HCl, pH 7.5). Bipartite transformations were prepared by adding 100 µg single stranded salmon sperm DNA and 1.5 to 5 µg each of the 5' and 3' DNA transformation fragments (3-10 µg total; see Table 2 for list of fragments) to approximately 200 µL protoplast mixture (10^{sup}.8 cells/mL). 1 mL 50% PEG in STC medium was then added to the salmon sperm DNA, transformation DNA, and protoplast mixture and the resulting combination was incubated for 15 minutes at room temperature.

Following incubation, recovery broth (0.4M sucrose, 1 g/L yeast extract, 1 g/L malt extract, 10 g/L glucose, pH 4.5) was added to the mixture and incubated at 27° C., 100 rpm, for 16 to 24 hours. Following the incubation, protoplasts were pelleted by centrifugation and resuspended in 1 mL PBS.

[0071] The resuspended protoplasts were plated on PDA+100 mg/L zeocin selection plates and incubated at 30-35° C. for at least 2-4 days until transformants grow. Resulting transformants were evaluated by colony PCR for integration of the indicated sequence. A PCR verified isolate was then designated as the indicated strain number. In some instances, more than one PCR verified isolate, e.g., “sister” isolates, are indicated by letters following the strain number. For example, strain 1-2 has 4 sister isolates, strains 1-2a, 1-2b, 1-2c, and 1-2d.

[0072] For example, Strain 1-1 was transformed with SEQ ID NO:12 and SEQ ID NO:13. SEQ ID NO: 12 contains i) 5' flanking DNA for targeted chromosomal integration into the ER3 locus (SEQ ID NO:24), and ii) a 5' portion of the Zeocin selectable marker (SEQ ID NO:66). SEQ ID NO:13 contains i) a 3' portion of the Zeocin selectable marker (SEQ ID NO:65), ii) the TEF2 terminator (SEQ ID NO:61), iii) an Mp6PGD promoter (SEQ ID NO:40), iv) a gene encoding the *Beauveria bassiana* ARD2DH homolog of SEQ ID NO:1, v) an Mp6PGD terminator (SEQ ID NO:51), and vi) a 3' ER3 flanking sequence (SEQ ID NO:25). Transformants were selected on PDA+100 mg/L zeocin selection plates and incubated at 30-35° C. for at least 2 days until transformants grow.

Resulting transformants were streaked for single colony isolation on PDA+zeocin plates and single colonies were selected. Selected colonies were evaluated by colony PCR for presence of the indicated sequence. PCR verified isolates were designated strains 1-2a, 1-2b, 1-2c, and 1-2d.

TABLE-US-00002

Strain	SEQ ID NO:	Organism	SEQ ID NO:	Marker	SEQ ID NO:	1-2a-d	1-1	13	<i>Beauveria bassiana</i>
1	Zeocin	12	1-3a-f	1-1	14	<i>Pichia stipitis</i>	2	Zeocin	12
1-4a-f	1-1	15	<i>Candida albicans</i>	3	Zeocin	12	1-5a-g	1-1	16
<i>Verticillium dahliae</i>	4	Zeocin	12	1-6a-f	1-1	17	<i>Fusarium proliferatum</i>	5	Zeocin
1-7a-e	1-1	18	<i>Aspergillus awamori</i>	6	Zeocin	12	1-8a-g	1-1	19
<i>Colletotrichum shioi</i>	7	Zeocin	12	1-					

9a-c 1-1 20 *Ophiostoma piceae* 8 Zeocin 12 1-10a-h 1-1 21 *Kwoniella heveanensis* 9 Zeocin 12 1-11a-f 1-1 22 *Pichia kudriavzevii* 10 Zeocin 12 1-12a-g 1-1 23 *Candida maltosa* 11 Zeocin 12

Example 3—Shake Flask Fermentation Assay

[0073] Strains 1-1, 1-2a-d, 1-3a-f, 1-4a-f, 1-5a-g, 1-6a-f, and 1-7a-e (outlined in Table 2 above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, arabitol, and ethanol production and glucose consumption. As indicated in the tables below, some strains were run in duplicate.

[0074] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0075] A 250 ml non-baffled flask containing production medium (Table 3) and antifoam CF-32 was inoculated with the seed culture to form the production culture with a starting OD600 of about 0.4 (approximately 0.8 mL of the seed culture). The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation. Samples were analyzed for glucose, ribitol, xylitol, erythritol, glycerol, arabitol, and ethanol by high performance liquid chromatography with refractive index detector. Fermentation results are reported in Tables C and D and FIGS. 3 and 4. The expression of either the *Beauveria bassiana* ARD2DH homolog (SEQ ID NO:1), the *Pichia stipitis* ARD2DH homolog (SEQ ID NO:2), or the *Candida albicans* ARD2DH homolog (SEQ ID NO:3) in *Moniliella pollinis* resulted in production of arabitol at levels above that produced in the parent strain 1-1.

TABLE-US-00003 TABLE 3 Production Medium Component Concentration (units) Glucose 300 (g/L) KH₂PO₄ 1.27 (g/L) (NH₄)₂HPO₄ 0.13 (g/L) (NH₄)₂SO₄ 1.80 (g/L) Urea 2.85 (g/L) Citric Acid 200.0 (mg/L) MgSO₄—7H₂O 690.2 (mg/L) FeSO₄—7H₂O 32.35 (mg/L) ZnSO₄—7H₂O 4.40 (mg/L) MnSO₄—H₂O 0.53 (mg/L) CuSO₄—5H₂O 0.45 (mg/L) CaCl₂—2H₂O 91.60 (mg/L) Na₂SO₄ 154.50 (mg/L) Thiamine-HCl 10 (mg/L) Choline-Cl 100 (mg/L) Antifoam CF-32 1.00 (g/L)

TABLE-US-00004 TABLE 4 Shake Flask Results from HPLC analysis Time Fermentation Broth Analyte (g/L) Strain point Glucose Ribitol Xylitol Erythritol Glycerol Ethanol 1-2a 72 h 138.974 0.952 n.a. 42.458 36.162 12.128 1-2a 72 h 154.818 0.877 n.a. 39.143 35.189 12.830 1-2b 72 h 147.230 0.858 n.a. 40.215 31.736 10.654 1-2c 72 h 171.870 1.066 n.a. 40.236 21.374 9.887 1-2d 72 h 123.766 0.840 n.a. 55.664 33.503 16.918 1-3a 72 h 53.969 1.232 n.a. 64.291 74.409 10.532 1-3b 72 h 197.030 0.702 n.a. 28.731 27.809 6.502 1-3b 72 h 187.903 0.560 n.a. 31.747 31.654 4.260 1-3c 72 h 130.937 0.997 n.a. 50.702 37.837 16.380 1-3d 72 h 142.550 0.999 n.a. 42.669 30.431 13.946 1-3d 72 h 96.848 1.325 n.a. 64.914 39.370 14.251 1-3e 72 h 73.658 1.242 n.a. 66.444 47.364 17.622 1-3f 72 h 121.154 1.021 n.a. 48.641 31.506 16.467 1-4a 72 h 151.311 0.651 n.a. 39.050 37.208 11.610 1-4a 72 h 156.634 0.473 n.a. 37.185 34.785 13.796 1-4b 72 h 85.931 0.814 n.a. 59.658 56.796 12.817 1-4b 72 h 97.442 0.715 n.a. 56.453 44.355 17.953 1-4c 72 h 94.773 0.941 n.a. 61.601 46.229 13.104 1-4d 72 h 111.870 0.832 n.a. 55.710 37.555 17.192 1-4d 72 h 139.961 0.941 n.a. 49.182 36.785 14.520 1-4e 72 h 172.284 0.516 n.a. 32.121 28.667 10.100 1-4f 72 h 74.533 1.349 n.a. 59.577 62.783 13.522 1-5a 72 h 202.443 0.596 0.056 23.629 14.945 14.594 1-5a 72 h 187.870 0.618 0.051 26.423 15.886 16.484 1-5b 72 h 112.036 1.256 0.049 57.326 36.291 13.694 1-5c 72 h 87.601 1.267 0.051 66.859 44.340 13.306 1-5d 72 h 144.311 1.036 0.024 45.567 30.897 16.153 1-5e 72 h 169.817 0.623 0.032 32.422 31.399 13.399 1-5f 72 h 111.758 1.115 0.037 53.894 35.848 14.645 1-5g 72 h 62.216 2.007 0.078 64.188 68.604 11.441 1-6a 72 h 95.951 0.841 0.045 60.461 35.540 18.394 1-6b 72 h 161.823 0.867 0.053 46.684 19.524 13.461 1-6c 72 h

333.753 0.036 n.a. 1.621 0.988 2.392 1-6d 72 h 60.287 1.324 n.a. 72.479 76.702 6.809 1-6d 72 h
 72.865 1.176 n.a. 67.559 58.915 12.283 1-6e 72 h 165.974 0.863 0.062 40.876 17.582 17.366 1-6f
 72 h 146.021 10.838 0.046 43.472 30.099 13.857 1-7a 72 h 175.392 0.689 0.024 34.772 23.845
 12.074 1-7b 72 h 142.213 0.739 0.041 42.201 23.216 20.619 1-7d 72 h 126.273 1.435 0.038 50.768
 36.879 15.641 1-7e 72 h 41.556 2.501 0.065 79.599 70.089 11.542 1-7e 72 h 32.295 2.501 0.057
 76.289 74.286 10.538 1-7f 72 h 213.204 0.704 n.a. 27.133 22.063 11.927 1-1 72 h 79.170 1.231
 0.040 65.958 47.596 12.070 1-1 72 h 97.698 0.991 n.a. 59.115 35.557 18.380

TABLE-US-00005 TABLE 5 Shake Flask Results from HPIC Analysis Time Analyte (g/L) Time
 Analyte (g/L) Strain Point Xylitol Ribitol Arabitol Strain Point Xylitol Ribitol Arabitol 1-2a 72 h
 n.a. 1.115 0.052 1-5e 72 h n.a. 0.857 0.039 1-2a 72 h n.a. 2.100 0.074 1-5f 72 h n.a. 1.370 0.052 1-
 2b 72 h n.a. 1.135 0.048 1-5g 72 h n.a. 1.934 0.069 1-2c 72 h n.a. 1.324 0.042 1-6a 72 h n.a. 0.966
 0.055 1-2d 72 h n.a. 0.988 0.055 1-6b 72 h n.a. 0.905 0.046 1-3a 72 h n.a. 1.424 0.759 1-6c 72 h
 n.a. 0.043 n.a. 1-3b 72 h n.a. 0.724 0.209 1-6d 72 h n.a. 1.574 0.067 1-3b 72 h n.a. 0.694 0.201 1-
 6d 72 h n.a. 1.288 0.056 1-3c 72 h n.a. 1.205 0.611 1-6e 72 h n.a. 1.183 0.051 1-3d 72 h n.a. 1.293
 0.465 1-6f 72 h n.a. 1.009 0.047 1-3d 72 h n.a. 1.589 0.659 1-7a 72 h n.a. 0.854 0.049 1-3e 72 h
 n.a. 1.639 0.708 1-7b 72 h n.a. 0.992 0.052 1-3f 72 h n.a. 1.189 0.596 1-7d 72 h n.a. 1.629 0.051 1-
 4a 72 h n.a. 0.750 0.211 1-7e 72 h n.a. 2.591 0.072 1-4a 72 h n.a. 0.750 0.212 1-7e 72 h n.a. 2.587
 0.071 1-4b 72 h n.a. 0.993 0.273 1-7f 72 h n.a. 0.706 0.049 1-4b 72 h n.a. 0.898 0.264 1-1 72 h n.a.
 1.292 0.058 1-4c 72 h n.a. 1.034 0.436 1-1 72 h n.a. 1.097 0.053 1-4d 72 h n.a. 0.930 0.287 1-2a 96
 h n.a. 1.941 0.137 1-4d 72 h n.a. 0.997 0.297 1-2a 96 h n.a. 1.941 0.312 1-4e 72 h n.a. 0.832 0.336
 1-2b 96 h n.a. 2.288 0.055 1-4f 72 h n.a. 1.494 0.575 1-2c 96 h n.a. 2.670 0.047 1-5a 72 h n.a.
 1.227 0.054 1-2d 96 h n.a. 1.476 0.156 1-5a 72 h n.a. 1.293 0.052 1-3a 96 h n.a. 1.529 0.822 1-5b
 72 h n.a. 1.317 0.053 1-3b 96 h n.a. 1.739 0.509 1-5c 72 h n.a. 1.321 0.057 1-3b 96 h n.a. 1.617
 0.397 1-5d 72 h n.a. 1.243 0.054 1-3c 96 h n.a. 1.893 1.124 1-3d 96 h n.a. 2.449 0.938 1-5e 96 h
 n.a. 1.558 0.011 1-3d 96 h n.a. 2.088 0.956 1-5f 96 h n.a. 2.227 0.032 1-3e 96 h n.a. 1.705 0.803 1-
 5g 96 h n.a. 2.126 0.062 1-3f 96 h n.a. 1.784 1.084 1-6a 96 h n.a. 1.185 0.065 1-4a 96 h n.a. 1.291
 0.397 1-6b 96 h n.a. 1.828 0.068 1-4a 96 h n.a. 1.341 0.395 1-6c 96 h n.a. 0.022 n.a. 1-4b 96 h n.a.
 1.170 0.335 1-6d 96 h n.a. 1.641 0.054 1-4b 96 h n.a. 1.039 0.314 1-6d 96 h n.a. 1.535 0.050 1-4c
 96 h n.a. 1.150 0.500 1-6e 96 h n.a. 2.283 0.068 1-4d 96 h n.a. 1.420 0.446 1-6f 96 h n.a. 1.795
 0.044 1-4d 96 h n.a. 1.775 0.525 1-7a 96 h n.a. 1.839 0.043 1-4e 96 h n.a. 1.618 0.619 1-7b 96 h
 n.a. 1.878 0.071 1-4f 96 h n.a. 1.691 0.709 1-7d 96 h n.a. 2.289 0.040 1-5a 96 h n.a. 1.730 0.013 1-
 7e 96 h n.a. 2.714 0.057 1-5a 96 h n.a. 1.636 0.015 1-7e 96 h n.a. 2.709 0.056 1-5b 96 h n.a. 1.853
 0.046 1-7f 96 h n.a. 1.468 0.059 1-5c 96 h n.a. 1.492 0.044 1-1 96 h n.a. 1.551 0.048 1-5d 96 h n.a.
 2.729 0.064 1-1 96 h n.a. 1.430 0.045

Example 4—Shake Flask Fermentation Assay

[0076] Strains 1-1, (outlined in Table 2 above), were run in shake flasks to assess xylitol, erythritol, ribitol, glycerol, arabitol, and ethanol production and glucose consumption. As indicated in the tables below, some strains were run in duplicate.

[0077] Strains were streaked out for biomass growth on YPD plates (bacteriological peptone 20 g/L, yeast extract 10 g/L, glucose 20 g/L, and agar 15 g/L) and incubated at 30° C. for 48-72 hours. Cells from the incubated YPD plates were scraped into 40 mL rich medium (170 g/L glucose, 10 g/L yeast extract) in a 250 mL baffled flask. Cells were incubated at 30° C. and 250 rpm until the optical density (OD600) reached 15-20 to form the seed culture. Optical density is measured at a wavelength of 600 nm with a 1 cm path length cuvette using a model Genesys20 spectrophotometer (Thermo Scientific). The seed culture reached an OD600 between 15-20 in about 32-50 hours.

[0078] A 250 ml non-baffled flask containing production medium (Table 3) and antifoam CF-32 was inoculated with the seed culture to form the production culture with a starting OD600 of about 0.4 (approximately 0.8 mL of the seed culture). The production culture was incubated at 35° C. and 250 rpm. Samples were taken from the production culture after 72 and 96 hours of incubation.

TABLE-US-00007 TABLE 7 Shake Flask Results from HPLC Analysis Fermentation Broth Analyte (g/L)

Strain	Time Point	Xylitol	Ribitol	Arabitol
1-8a	72 h	n.a.	0.359	0.074
1-8b	72 h	0.214	0.869	0.109
1-8c	72 h	0.272	0.832	0.102
1-8d	72 h	0.230	0.761	0.096
1-8e	72 h	n.a.	1.037	0.094
1-8f	72 h	0.327	0.874	0.109
1-8g	72 h	0.288	0.589	0.094
1-9a	72 h	n.a.	0.915	0.098
1-9b	72 h	0.169	0.723	0.085
1-9c	72 h	n.a.	1.112	0.120
1-9d	72 h	0.132	0.747	0.031
1-9e	72 h	0.135	0.769	0.031
1-9f	72 h	n.a.	1.367	0.091
1-9g	72 h	n.a.	1.959	0.108
1-10a	72 h	n.a.	0.587	0.286
1-10b	72 h	n.a.	1.078	0.803
1-10c	72 h	n.a.	1.046	0.833
1-10d	72 h	0.186	1.038	0.519
1-10e	72 h	n.a.	1.003	0.623
1-10f	72 h	n.a.	1.754	0.091
1-10g	72 h	n.a.	1.016	0.816
1-10h	72 h	n.a.	0.181	1.108
1-10i	72 h	n.a.	0.137	1.108
1-11a	72 h	n.a.	0.955	0.319
1-11b	72 h	n.a.	1.090	0.105
1-11c	72 h	0.288	0.804	0.102
1-11d	72 h	0.160	0.567	0.066
1-11e	72 h	n.a.	0.955	0.137
1-11f	72 h	n.a.	1.029	0.110
1-11g	72 h	n.a.	0.391	0.057
1-12a	72 h	0.289	0.846	0.615
1-12b	72 h	0.252	0.762	0.446
1-12c	72 h	0.271	0.909	0.701
1-12d	72 h	0.273	1.110	0.426
1-12e	72 h	0.277	0.849	0.653
1-12f	72 h	0.317	1.001	0.724
1-12g	72 h	n.a.	0.722	0.517
1-1	72 h	n.a.	1.216	0.126
1-8a	96 h	0.212	1.317	0.113
1-8b	96 h	0.264	1.604	0.116
1-8c	96 h	0.319	1.408	0.092
1-8d	96 h	0.289	1.416	0.090
1-8e	96 h	n.a.	1.997	0.098
1-8f	96 h	0.295	1.122	0.081
1-8g	96 h	0.345	0.896	0.073
1-9a	96 h	0.268	1.894	0.108
1-9b	96 h	0.180	1.621	0.145
1-9c	96 h	n.a.	1.751	0.126
1-9d	96 h	0.114	1.642	n.a.
1-9e	96 h	0.125	1.596	n.a.
1-9f	96 h	0.296	3.095	0.060
1-9g	96 h	0.231	3.749	0.189
1-10a	96 h	0.277	0.942	0.573
1-10b	96 h	0.269	1.667	1.311
1-10c	96 h	0.270	1.535	1.198
1-10d	96 h	0.451	2.210	1.248
1-10e	96 h	0.347	2.260	1.228
1-10f	96 h	0.240	3.247	0.081
1-10g	96 h	0.331	1.541	1.171
1-10h	96 h	0.311	2.391	0.178
1-10i	96 h	n.a.	1.539	0.464
1-11a	96 h	n.a.	2.099	0.131
1-11b	96 h	0.388	1.206	0.082
1-11c	96 h	0.236	1.334	0.119
1-11d	96 h	n.a.	1.397	0.080
1-11e	96 h	n.a.	1.328	0.082
1-11f	96 h	0.225	1.259	0.029
1-12a	96 h	0.505	1.676	1.324
1-12b	96 h	0.516	1.670	1.209
1-12c	96 h	0.509	1.747	1.542
1-12d	96 h	0.289	1.759	0.688
1-12e	96 h	0.501	1.605	1.523
1-12f	96 h	0.528	1.798	1.535
1-12g	96 h	0.381	1.636	1.382
1-1	96 h	0.526	1.491	1.383
1-1	96 h	n.a.	0.257	1.509
1-1	96 h	n.a.	0.096	1.866
1-1	96 h	n.a.	0.090	

Claims

1. A genetically engineered yeast cell capable of producing arabitol, the engineered yeast cell comprising: an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 80% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.
2. The yeast cell of claim 1, wherein the yeast cell is an osmotolerant yeast cell.
3. The yeast cell of claim 1, wherein the yeast cell is a cell of the subphylum Ustilaginomycotina.
4. The yeast cell of claim 1, wherein the yeast cell is selected from the group consisting of *Trichosporonoides megachiliensis*, *Trychosporonoides oedocephalis*, *Trychosporonoides nigrescens*, *Pseudozyma tsukubaensis*, *Trigonopsis variabilis*, *Moniliella*, *Ustilaginomycetes*, *Trichosporon*, *Yarrowia lipolytica*, *Penicillium*, *Torula*, *Pichia*, *Candida*, *Candida magnoliae*, and *Aureobasidium*
5. The yeast cell of claim 1, wherein the yeast cell is a yeast cell of the *Moniliella* genus.
6. A genetically engineered *Moniliella* cell capable of producing arabitol, the engineered *Moniliella* cell comprising: an exogenous polynucleotide sequence encoding an arabitol 2-dehydrogeanse (ARD2DH) enzyme comprising a sequence at least 85% identical to at least one of SEQ ID NOs:1, 2, 3, 9, or 11.
7. The yeast cell of claim 6, wherein the cell is a *Moniliella pollinis* cell.
8. The yeast cell of claim 1, wherein the yeast cell is capable of producing arabitol at a titer of at least 0.2 g/L when used in a fermentation process in the presence of dextrose at 35° C. for 96 hours.
9. (canceled)
10. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is integrated into the genome of the yeast cell at a loci selected from the ER1 locus, the ER3 locus, the PDC1 locus, the

pyrF locus, the TRP3 locus, the gpdIIA locus, and the gpdIIB locus.

11. The yeast cell of claim 1, wherein the exogenous polynucleotide sequence is operably linked to a heterologous or artificial promoter selected from the group consisting of pyruvate kinase 1 promoter (PYK1p; SEQ ID NO:49), 6-phosphogluconate dehydrogenase promoter (6PGDp; SEQ ID NO:40), glyceraldehyde-3-phosphate dehydrogenase promoter (TDH3p; SEQ ID NO:42), translational elongation factor 1 promoter (TEFp; SEQ ID NO:43), modified TEFp (SEQ ID NO:41), phosphoglucomutase 1 promoter (PGM1p; SEQ ID NO:44), 3-phosphoglycerate kinase promoter (PGK1p; SEQ ID NO:45), enolase promoter (ENO1p; SEQ ID NO:46), asparagine synthetase promoter (ASNSp; SEQ ID NO:47), 50S ribosomal protein L1 promoter (RPLAp; SEQ ID NO:48), and RPL16B (SEQ ID NO:50).

12. (canceled)

13. (canceled)

14. The yeast cell of claim 1, wherein the ARD2DH enzyme has a sequence at least 90% identical to SEQ ID NO:2, 3, 9, and/or 11.

15. The yeast cell of claim 6, wherein the ARD2DH enzyme has a sequence at least 90% identical to SEQ ID NO: 2, 3, 9, and/or 11.

16. A method for producing arabitol, the method comprising: contacting a substrate comprising dextrose with the engineered yeast cell of claim 1, wherein fermentation of the substrate by the engineered yeast produces arabitol.

17. A method for producing arabitol, the method comprising: contacting a substrate comprising dextrose with the engineered yeast cell of claim 6, wherein fermentation of the substrate by the engineered yeast produces arabitol.

18. (canceled)

19. The method of claim 16, wherein the fermentation temperature is at or between 25° C. to 45° C. and the volumetric oxygen uptake rate (OUR) is between 0.5 to 40 mmol O₂/(L.Math.h).

20. The method of claim 16, wherein erythritol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

21. The method of claim 16, wherein erythritol production is less than 60 g/L when the fermentation is run at 35° C. for 96 hours.

22. The method of claim 16, wherein arabitol production is at least 0.2 g/L when the fermentation is run at 35° C. for 96 hours.

23. The method of claim 16, wherein glycerol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.

24. The method of claim 16, wherein ethanol production is reduced relative to an equivalent fermentation run with an equivalent yeast cell lacking the exogenous polynucleotide sequence.
